DISK MANAGEMENT

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Basic Disk Management :

1. Partition
2. Formatting
3. Mounting

What is Disk - It is a physical component which is used to store digital data. We can not directly store data inside it.

What is Partition - It is the logical part of a physical disk.

What is partitioning - It is a process of dividing a physical disk into multiple logical disks.

What is Partition Style / Partition Scheme - it is the rule/process which will be used while partitioning.

What is File System - File System are the grammar or rules which will be used while accessing data from a drive.

Example of Linux File Systems - ext4,xfs,vfat,exfat.

MBR :-

1. Master Boot Record

2. Introduced by IBM
3. One Disk can have max 4 partition
4. Partitions can be of 2 types
    a. Primary : Used mainly to store operating systems. Also data can be stored.
    b. Extended : It can not store any kind of data directly. First we need to divide it into logical volumes then we can store data. In MBR partition style only 1 extended partition can be created.
5. MBR partition style scheme :
    a. 4 Primary Partition - Only Supported in Linux
    b. 3 Primary 1 Extended - Supported both in windows and linux (Default in Windows)
    c. 2 Primary 1 Extended - Only supported in Linux
    d. 1 Primary 1 Extended - Only Supported in Linux
6. Tool to use while performing partition for MBR : fdisk

GPT :

1. GUID Partition Table
2. Introduced by Intel
3. 128 partitions can be created
4. Best for advanced storage solutions

5. Tools to use : gdisk

To see summary information about the disks

```
[root@node1 ~]# lsblk
NAME          MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
sda             8:0    0   40G  0 disk
├─sda1          8:1    0    1G  0 part /boot
└─sda2          8:2    0   39G  0 part
  ├─rhel-root 253:0    0   37G  0 lvm  /
  └─rhel-swap 253:1    0    2G  0 lvm  [SWAP]
sdb             8:16   0   20G  0 disk
sr0            11:0    1 1024M  0 rom
[root@node1 ~]#
```

To see detailed information about disks

```
[root@node1 ~]# fdisk -l


Disk /dev/sda: 40 GiB, 42949672960 bytes, 83886080 sectors
Disk model: VBOX HARDDISK
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xfa697919
```

1. Added a Hard disk
2. Checked hdd is connected or not --> # lsblk # fdisk -l
3. for creating a partition with MBR Style
   # fdisk /dev/sdb

Press m for help
press n for a new partition
press partition type - Primary - P extended - E
Serial No - 1
First sector - press enter
last sector - +2G
press w to save and exit

4. format the partition using xfs file system
   # mkfs.xfs /dev/sdb1
5. update the partition table - # partprobe # udevadm settle

```
6. Mounting - 1. Temporary(/dir1)       2. Permanent(/dir2)
temp : # mount /dev/sdb1 /dir1
Perm : # vim /etc/fstab
go to last line
insert mode
/dev/sdb1   /dir2    xfs   defaults    0 0
save and exit
update the system's partition : # mount -a   # systemctl daemon-reload
```

Partition Style : GPT

Tool to use : gdisk

- For exam always install gdisk : # dnf install gdisk -y

Create a 5 GB Partition using gdisk utility

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```
[root@srv1 ~]# gdisk /dev/sdb
```

```
Command (? for help): n
Partition number (1-128, default 1): 1
First sector (34-41943006, default = 2048) or {+-}size{KMGTP}:
Last sector (2048-41943006, default = 41943006) or {+-}size{KMGTP}: +5G
Current type is 8300 (Linux filesystem)
Hex code or GUID (L to show codes, Enter = 8300):
Changed type of partition to 'Linux filesystem'

Command (? for help): w

Final checks complete. About to write GPT data. THIS WILL OVERWRITE EXISTING
PARTITIONS!!

Do you want to proceed? (Y/N): y
OK; writing new GUID partition table (GPT) to /dev/sdb.
```

Updating the partition table

```
[root@srv1 ~]# udevadm settle
```

Formatting the drive

```
[root@srv1 ~]# mkfs.xfs /dev/sdb1
```

Creating mount points

```
[root@srv1 ~]# mkdir /dir1
[root@srv1 ~]#
[root@srv1 ~]# mkdir /dir2
```

Temporary mounting

```
[root@srv1 ~]# mount /dev/sdb1 /dir1
```

Permanent mounting

```
[root@srv1 ~]# vim /etc/fstab
```

fstab file entry for permanent mounting

```
/dev/sdb1                /dir2                    xfs     defaults        0 0
~
```

Force the system to read fstab file without restarting

```
[root@srv1 ~]# mount -a
mount: (hint) your fstab has been modified, but systemd still uses
       the old version; use 'systemctl daemon-reload' to reload.
[root@srv1 ~]#
```

Or

```
[root@srv1 ~]# systemctl daemon-reload
```

To see the mount points

```
[root@srv1 ~]# df -Th
Filesystem             Type      Size  Used Avail Use% Mounted on
devtmpfs               devtmpfs  4.0M     0  4.0M   0% /dev
tmpfs                  tmpfs     871M     0  871M   0% /dev/shm
tmpfs                  tmpfs     349M  5.2M  344M   2% /run
efivarfs               efivarfs  256K   56K  196K  23% /sys/firmware/efi/efivars
/dev/mapper/rhel-root  xfs        37G  1.9G   35G   6% /
/dev/sda2              xfs       960M  309M  652M  33% /boot
/dev/sda1              vfat      599M  7.1M  592M   2% /boot/efi
tmpfs                  tmpfs     175M     0  175M   0% /run/user/0
tmpfs                  tmpfs     175M     0  175M   0% /run/user/1234
/dev/sdb1              xfs       5.0G   68M  4.9G   2% /dir1
[root@srv1 ~]#
```

Or to filter exact drive details

```
[root@srv1 ~]# df -Th | grep /dev/sdb1
/dev/sdb1              xfs       5.0G   68M  4.9G   2% /dir1
[root@srv1 ~]#
```

fstab file entry details

==========

```
/dev/sdb1                /dir2                    xfs     defaults        0 0
~
```

1. /dev/sdb1 : drive identification , simply drive name , in this column we can use drive name , drive label and drive UUID
2. /dir2 : permanent mount point of the drive, nature persistent. [persistent : available after restart]
3. xfs : file system which is used to format the drive
4. defaults : function of the drive,if it will be used to store data then enter defaults or if we want to apply advanced storage solutions like RAID,Quota,Data Deduplication,Storage Pool,Iscsi we need to mention here.
5. 0 : it is known as Read write dump : it is an activity that the system performs during booting. If the value of this field is 1 then during booting the system will enter some values in the fstab file to check if the file is in read and write mode or not. System can do this activity by changing the timestamp of the system. If the system is unable to write anything in the file system understands that there are some serious problems regarding disk management and the system will start in recovery mode.

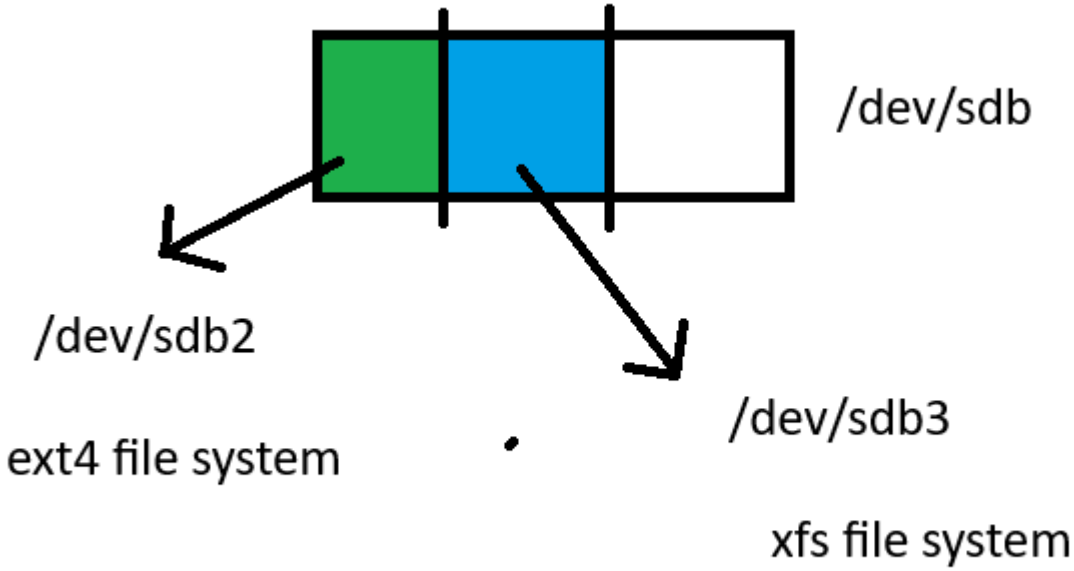
6. 0 : this field denotes file system checking. In this field we can put 3 values 0/1/2 . 0 means no checking , 1 means this drive will be checked first and 2 means system OS drives will be checked first then this drive will be checked.

Mounting Types :

1. Drive Mounting : mounting using the drive id . example : /dev/sdb1

2. Label Mounting : one label will be given to the drive. Example : softwares
3. UUID Mounting :

Label Mounting



Set a label for ext4 drive

```
[root@srv1 ~]# e2label /dev/sdb2 softwares
[root@srv1 ~]#
```

Set a label for xfs drive

```
[root@srv1 ~]# xfs_admin -L datastore /dev/sdb3
writing all SBs
new label = "datastore"
[root@srv1 ~]#
```

Temp Label Mounting

```
[root@srv1 ~]# # dir10 : for temp label mounting of /dev/sdb2 which is formatted with ext4 filesystem
[root@srv1 ~]#
[root@srv1 ~]# mount LABEL=softwares /dir10
[root@srv1 ~]#
[root@srv1 ~]# df -Th | grep /dir10
/dev/sdb2              ext4      2.9G   24K  2.8G   1% /dir10
[root@srv1 ~]#
```

Perm Label Mounting

```
[root@srv1 ~]# vim /etc/fstab
```

```
LABEL=softwares          /dir11                    ext4     defaults        0 0
~
```

Update the system for changing fstab file

```
[root@srv1 ~]# systemctl daemon-reload
[root@srv1 ~]#
```

UUID Mounting :

1. Full form Universally Unique Identifier
2. It is a 128 bit binary character
3. 32 bit hexadecimal digits
4. When we create and format a drive system allocates a unique id for each drives

Creating partition for a nvme hard disk

```
[root@srv1 ~]# gdisk /dev/nvme0n1
GPT fdisk (gdisk) version 1.0.7

Partition table scan:
  MBR: protective
  BSD: not present
  APM: not present
  GPT: present

Found valid GPT with protective MBR; using GPT.

Command (? for help): n
Partition number (2-128, default 2): 2
First sector (34-41943006, default = 4196352) or {+-}size{KMGTP}:
Last sector (4196352-41943006, default = 41943006) or {+-}size{KMGTP}: +2G
Current type is 8300 (Linux filesystem)
Hex code or GUID (L to show codes, Enter = 8300):
Changed type of partition to 'Linux filesystem'

Command (? for help): w

Final checks complete. About to write GPT data. THIS WILL OVERWRITE EXISTING
PARTITIONS!!

Do you want to proceed? (Y/N): y
OK; writing new GUID partition table (GPT) to /dev/nvme0n1.
The operation has completed successfully.
[root@srv1 ~]#
```

Update partition table

```
[root@srv1 ~]# udevadm settle
[root@srv1 ~]#
```

Formatting the drive using ext4 file system

```
[root@srv1 ~]# mkfs.ext4 /dev/nvme0n1p1
```

Formatting the drive using xfs file system

```
[root@srv1 ~]# mkfs.xfs /dev/nvme0n1p2
```

Find UUID for the drive

```
[root@srv1 ~]# blkid | grep /dev/nvme0n1p1
/dev/nvme0n1p1: UUID="d6cb9b12-7a69-413f-807a-28c8a9ff9a5a" TYPE="ext4" PARTLABEL="Linux filesystem" PARTUUID="6b91c3cb-4afa-4ce
c-bd53-1e3569d51774"
```

Temporary UUID Mounting

```
[root@srv1 ~]# mount UUID="d6cb9b12-7a69-413f-807a-28c8a9ff9a5a" /dir12
[root@srv1 ~]#
```

Permanent Mounting using UUID

```
UUID="d6cb9b12-7a69-413f-807a-28c8a9ff9a5a"     /dir13  xfs     defaults        0 0
```

Update the fstab file

```
[root@srv1 ~]# systemctl daemon-reload
[root@srv1 ~]#
```